House Price Prediction – Project Summary

Objective

The objective of this project was to build machine learning models that can predict house prices based on property features such as area, number of bedrooms, bathrooms, stories, parking availability, location preferences, and furnishing status.

Data Preparation

The Housing dataset was loaded and explored using Pandas. Missing values and duplicate records were checked and handled appropriately. Categorical variables such as main road access, guest room availability, air conditioning, and furnishing status were converted into numerical form using encoding techniques to prepare the data for machine learning.

Model Development

Two regression models were developed:

1. **Linear Regression**
2. **Random Forest Regressor**

The dataset was split into training and testing sets using an 80:20 ratio. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Key Findings

- Property area was the most important factor influencing house prices.
- Bathrooms, bedrooms, stories, parking spaces, and preferred area also contributed significantly to price prediction.
- Both models achieved good predictive performance, with Random Forest generally capturing complex relationships more effectively.
- The dataset showed a right-skewed distribution, indicating the presence of a few high-priced properties.

Recommendation

Real estate businesses should focus primarily on property size, location, and key structural features when estimating property values and developing marketing strategies. These factors have the strongest impact on house prices and can help improve pricing accuracy.

Conclusion

This project successfully demonstrated how machine learning techniques can be used to predict house prices and identify the factors that drive property value. The results highlight the importance of data-driven decision-making in the real estate industry.